

HF Radio Amplifier with Lowpass/Bandpass Filters

Group 4

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Introduction

- BC wildfires have devastated rural areas and caused frequent communication blackouts [1].
- During disasters, power grid failures and cellular/internet outages severely limit coordination between first responders, agencies, and the public [2].
- VHF/UHF radios lack range and rely on costly, damage-prone repeaters [3, 4, 5].
- These systems don't work well across BC's vast, rugged interior [6].
- HF radio offers 100–300+ km range and works without towers or ISPs [7].
- It can be solar-powered, rapidly deployed, and stays online during outages [8].
- Despite previous success in disaster relief, HF communication isn't widely adopted due to high cost and bulky equipment [9, 10].
- Affordable low-power HF radios now exist but need an amplifier to reach full range [11].
- A compact, portable HF amplifier can close this gap and keep communities connected .

Design Objectives

- Design and Evaluate an HF Radio Amplifier:** Develop a 1.8 – 30 MHz amplifier with integrated lowpass and bandpass filters for signal clarity and harmonic suppression.
- Deliver a Working Prototype:** Produce a functional amplifier prototype with operational filters.
- Investigate Feasibility:** Assess the potential of amateur radio infrastructure as a low-cost emergency communication solution.

Alternative Solutions

- Three popular portable commercial amplifiers HF amplifiers were analyzed to overview the price, features, and performance of each commercial option:

1. RM Italy KL203P [12]:

- 100W output | 109×125×35 mm | 325g | \$180 CAD
- No filtration, minimal cooling, no meter or tuner.



2. Xiegu XPA125B [13]:

- 100W output | 260×150×100 mm | 2.66 kg | \$830 CAD
- Has SWR/power meter and tuner.
- Lacks cooling features.



3. Elecraft KXPA100 [14]:

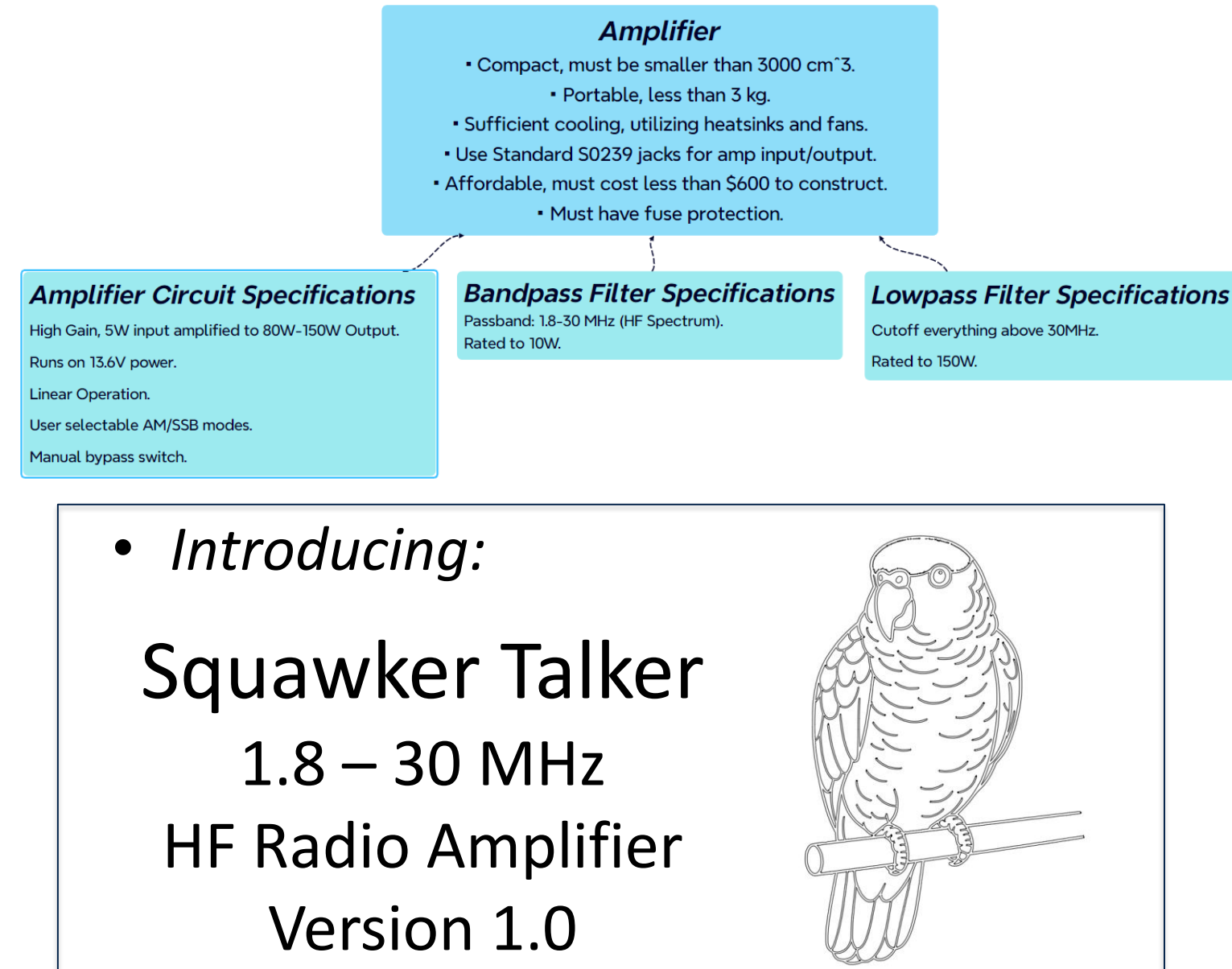
- 100W output | 254×130×108 mm | 3.2 kg | \$2270 CAD
- Only model with filtration and real cooling.
- Includes SWR/power meter, tuner, and bypass switch.



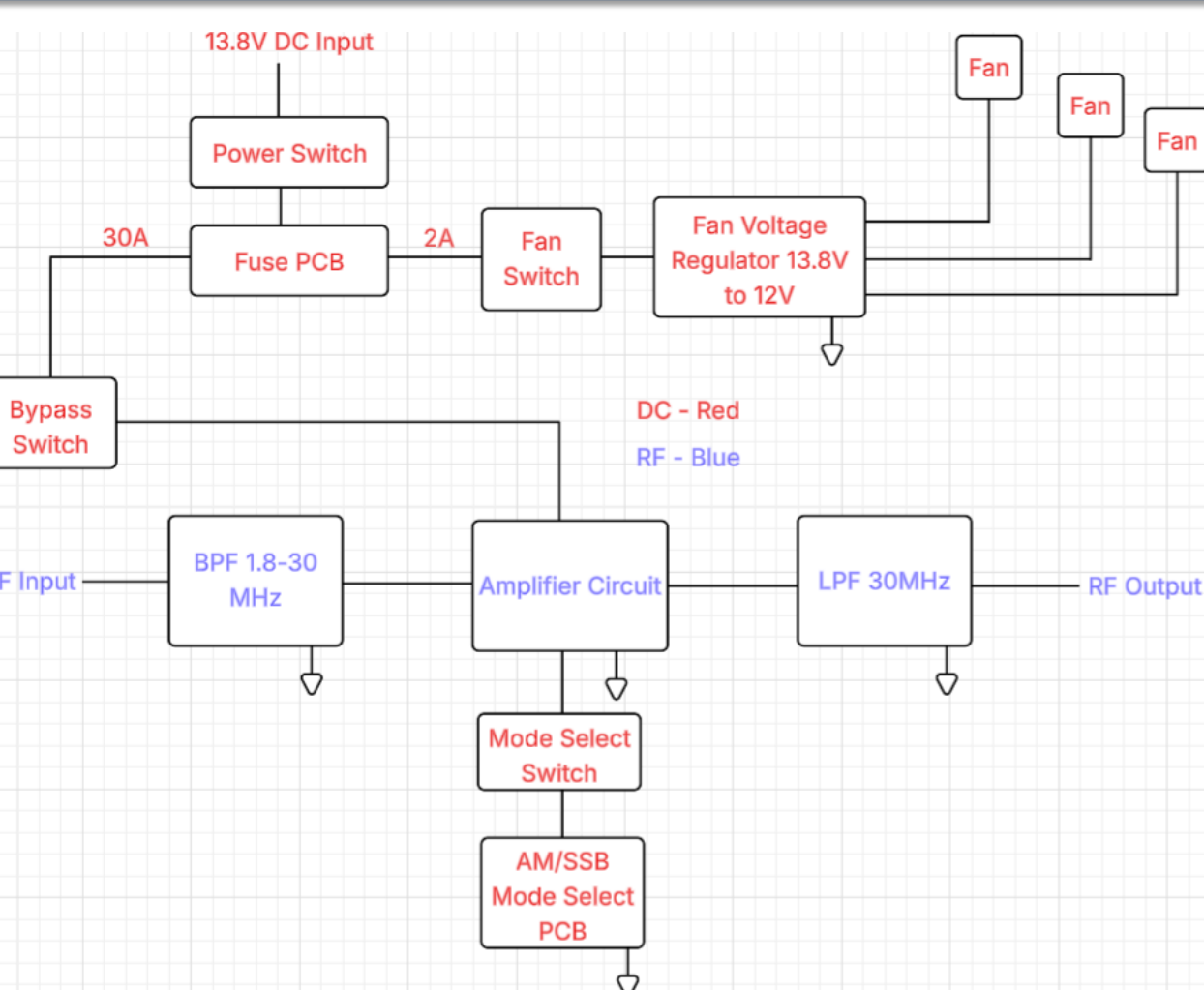
- Cheaper amp models skimp on filtration and thermal management, and only the premium, expensive Elecraft amplifier has these features. Therefore, a gap in the market for a mid cost amplifier with proper filtration and cooling was identified.

Feature Specifications

- Overviewing the alternative solutions, a set amplifier and subsystem features, and their individual specifications was created.

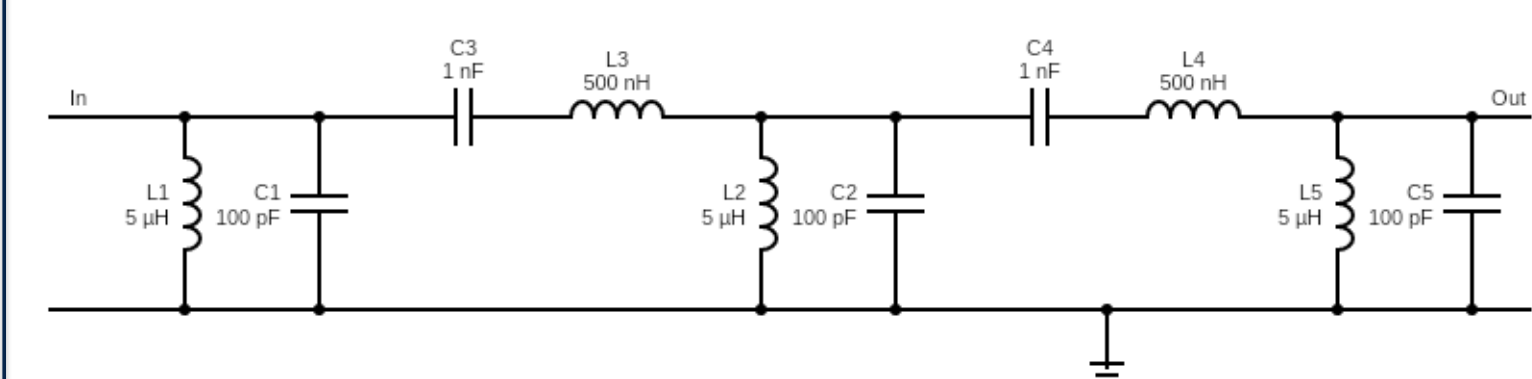


Design – Block Diagram

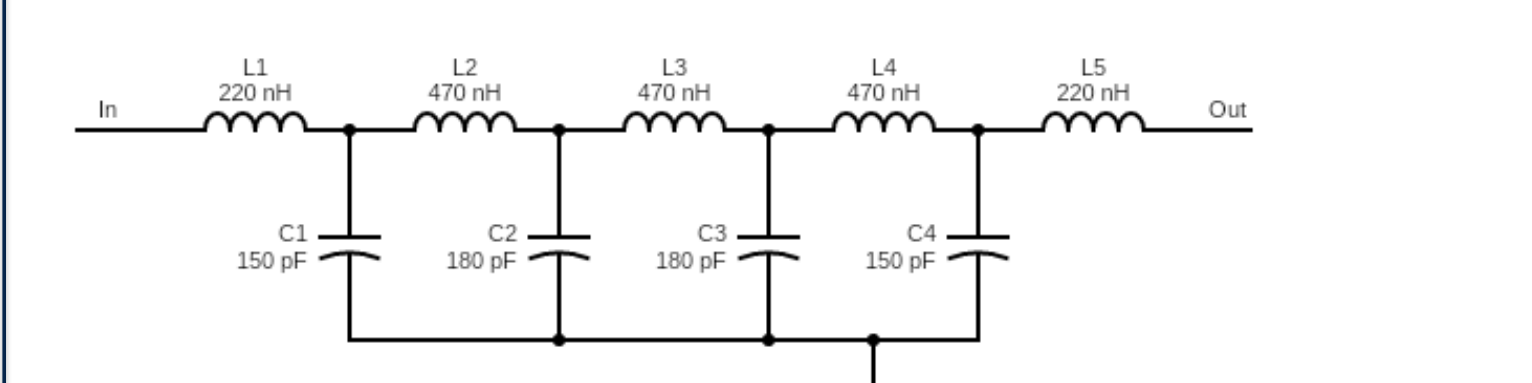


Design – Filter Circuits

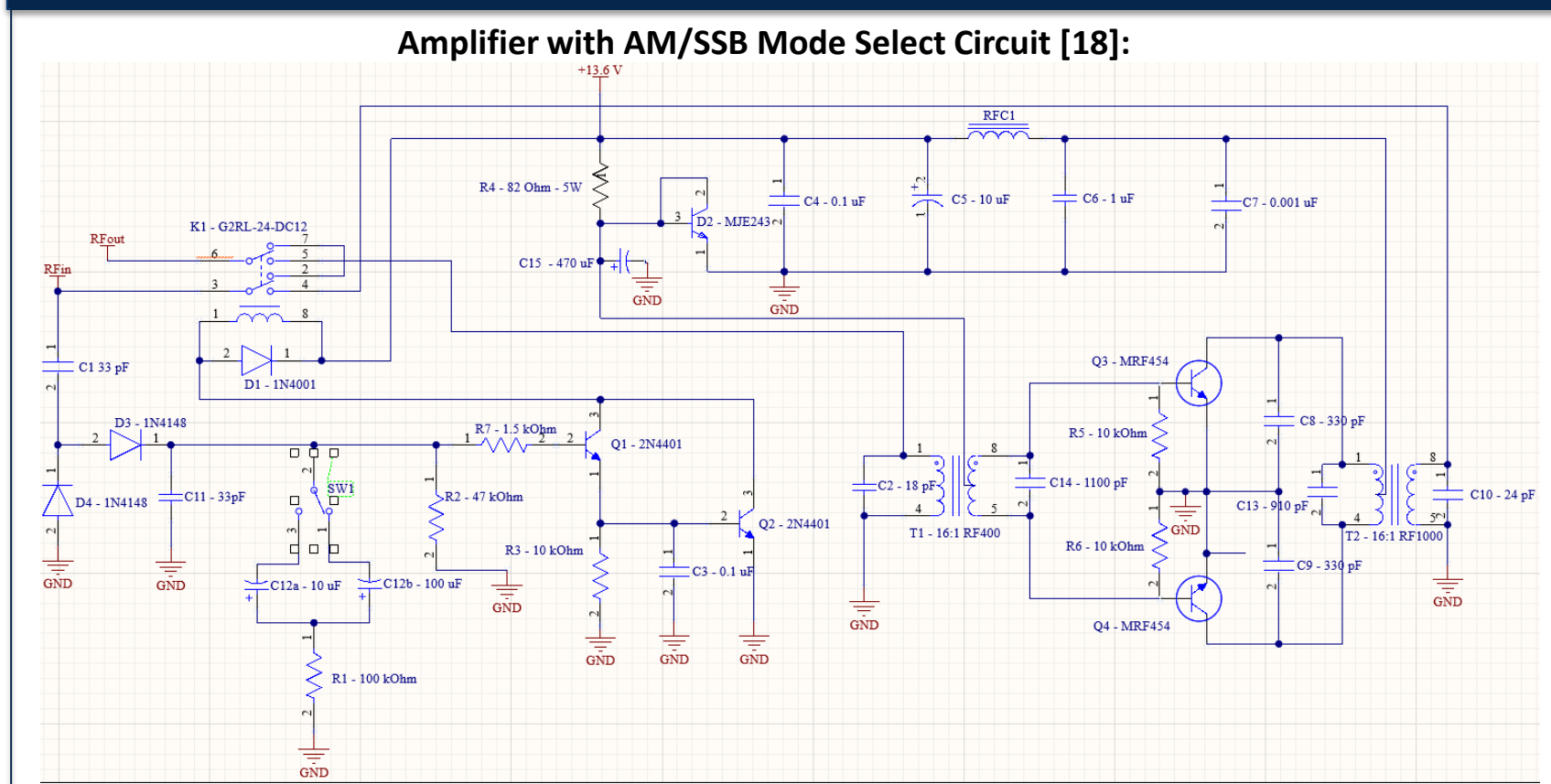
Constant-k Pi LC Network 1.8 – 30MHz Bandpass Filter [15, 16]:



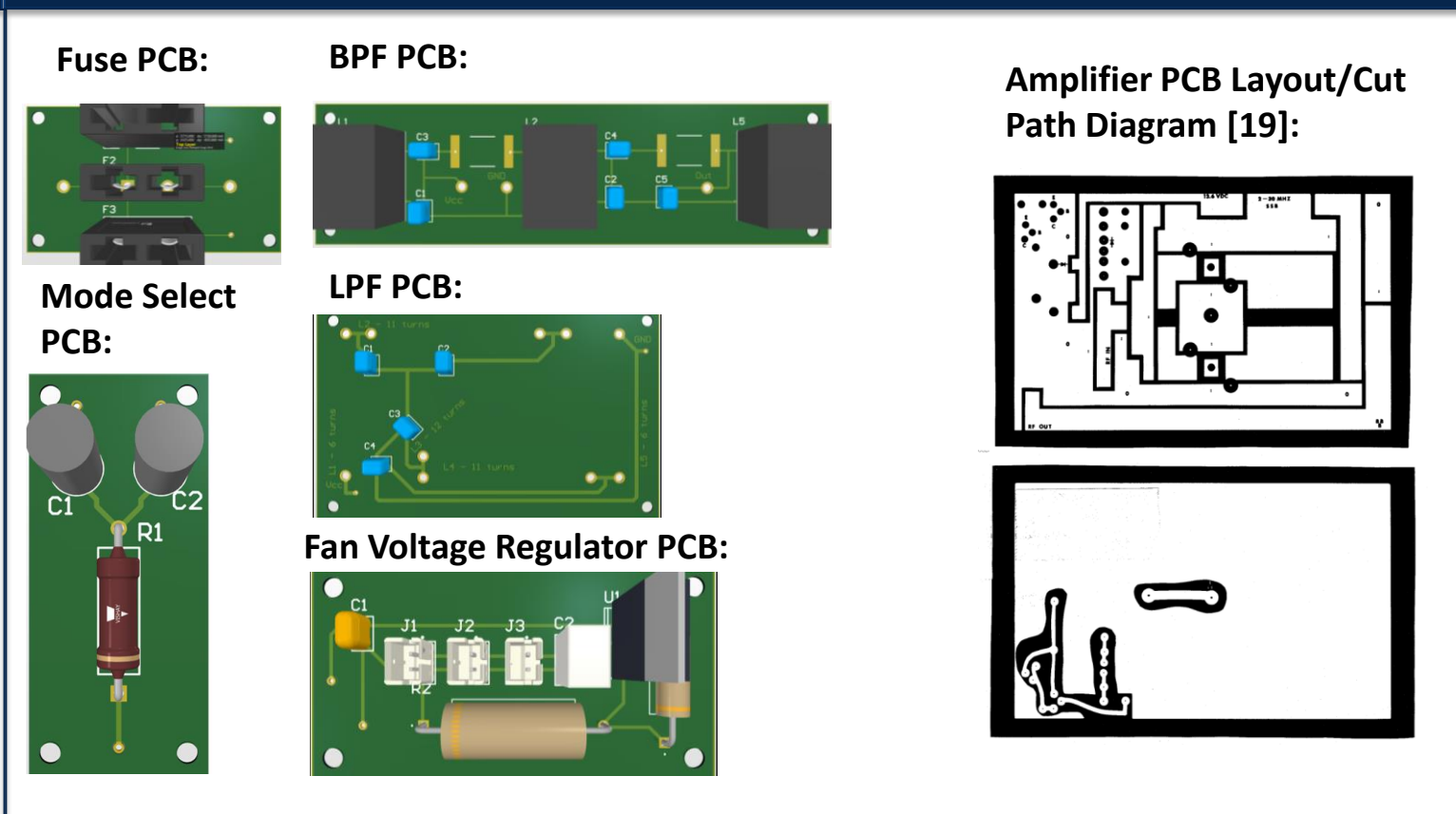
9th Order T base Chebyshev 30MHz Low Pass Filter [17]:



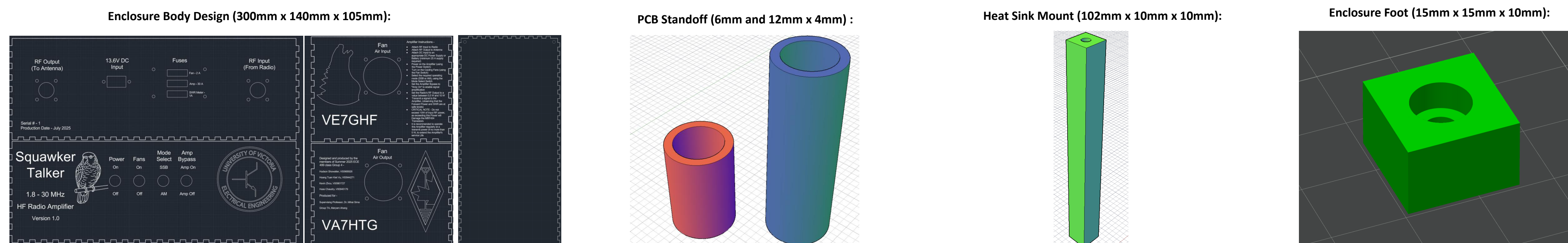
Design – Amplifier Circuit



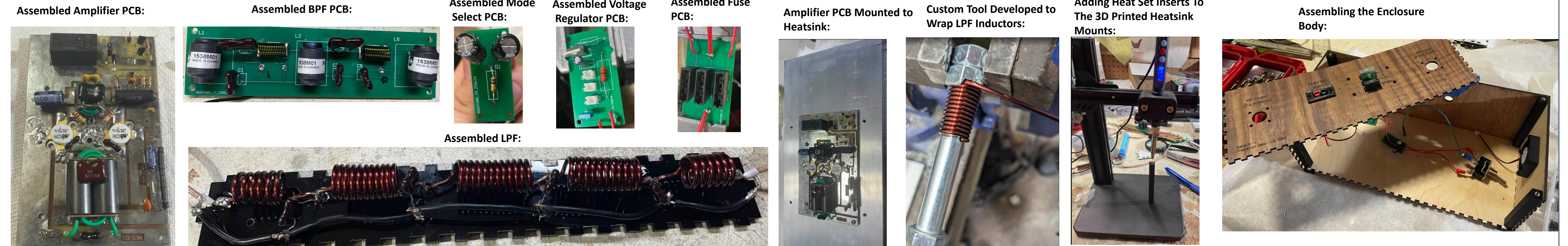
Design – PCB Layouts



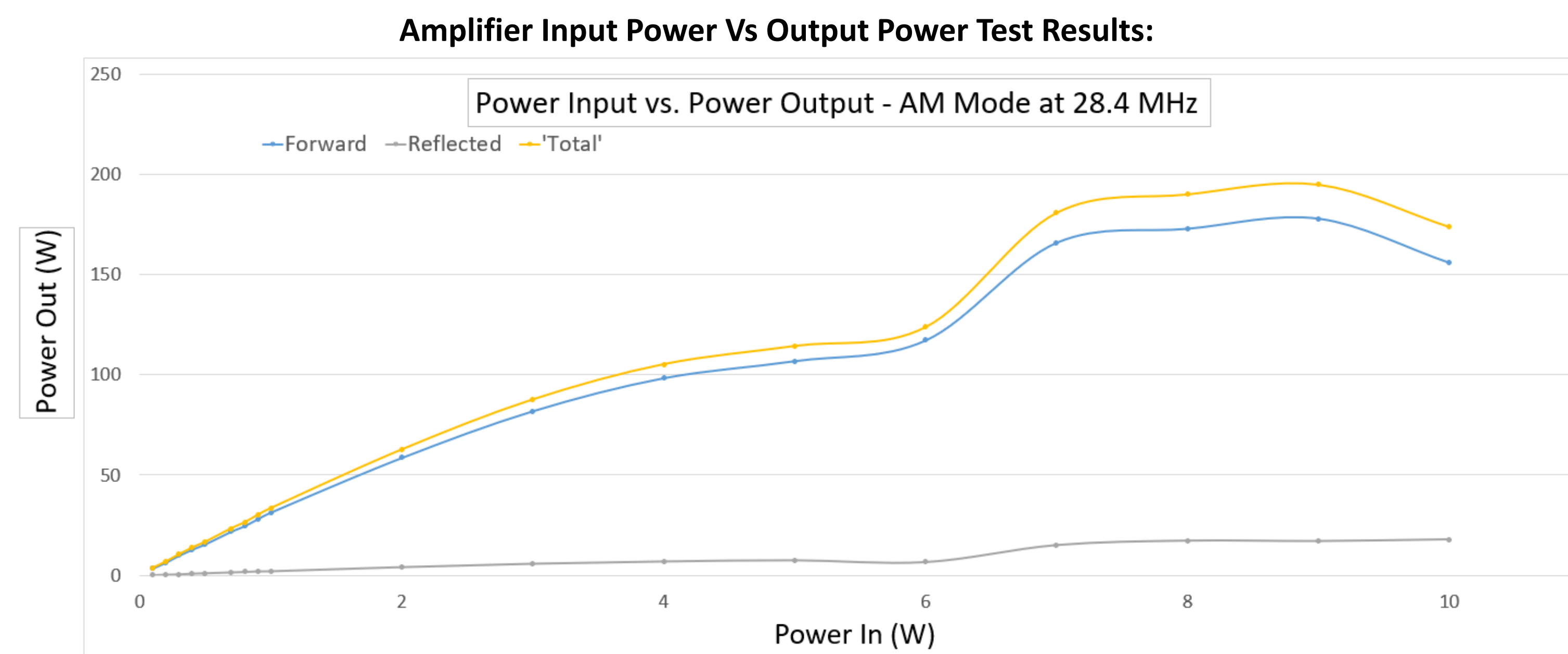
Design – Enclosure



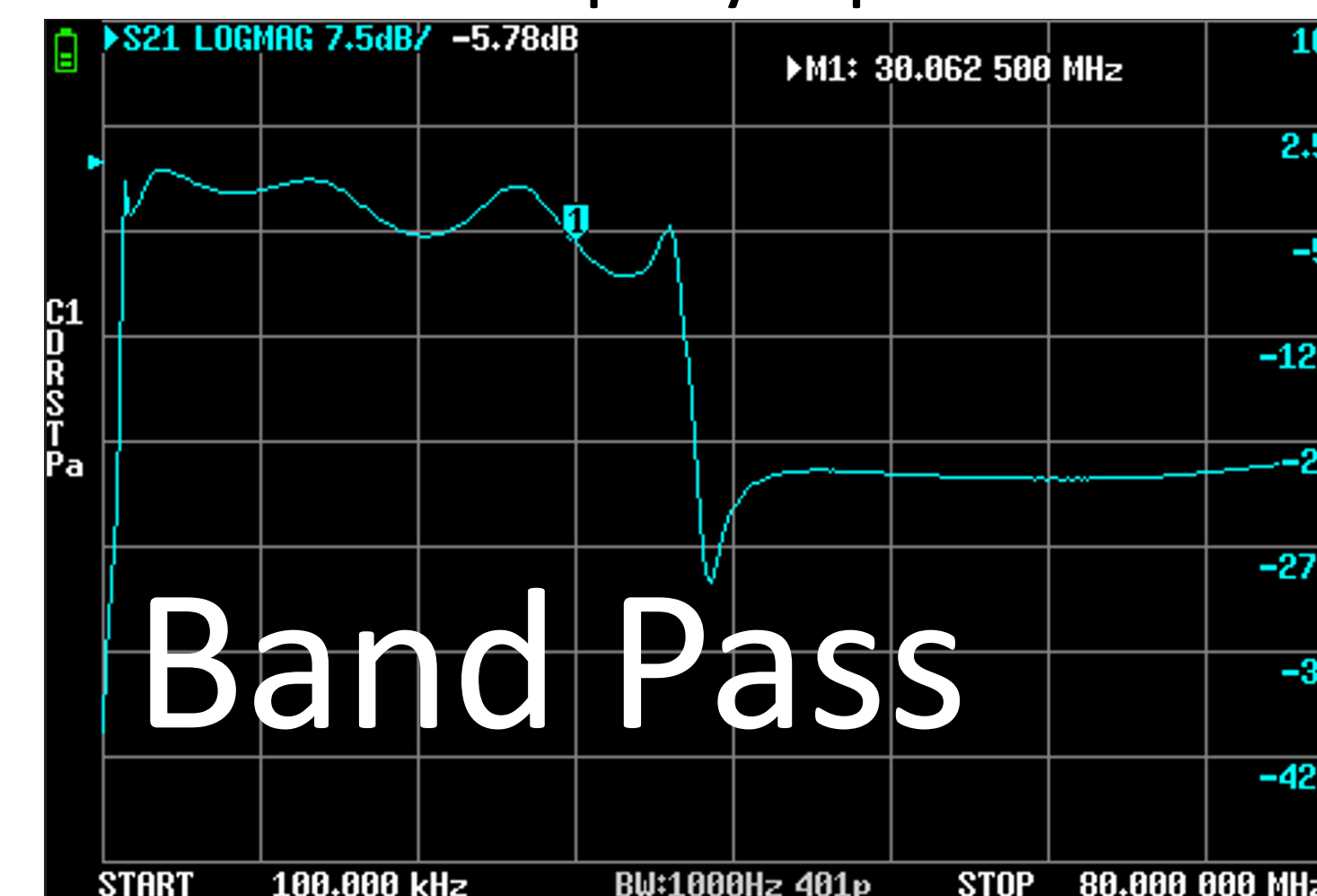
Prototype Assembly



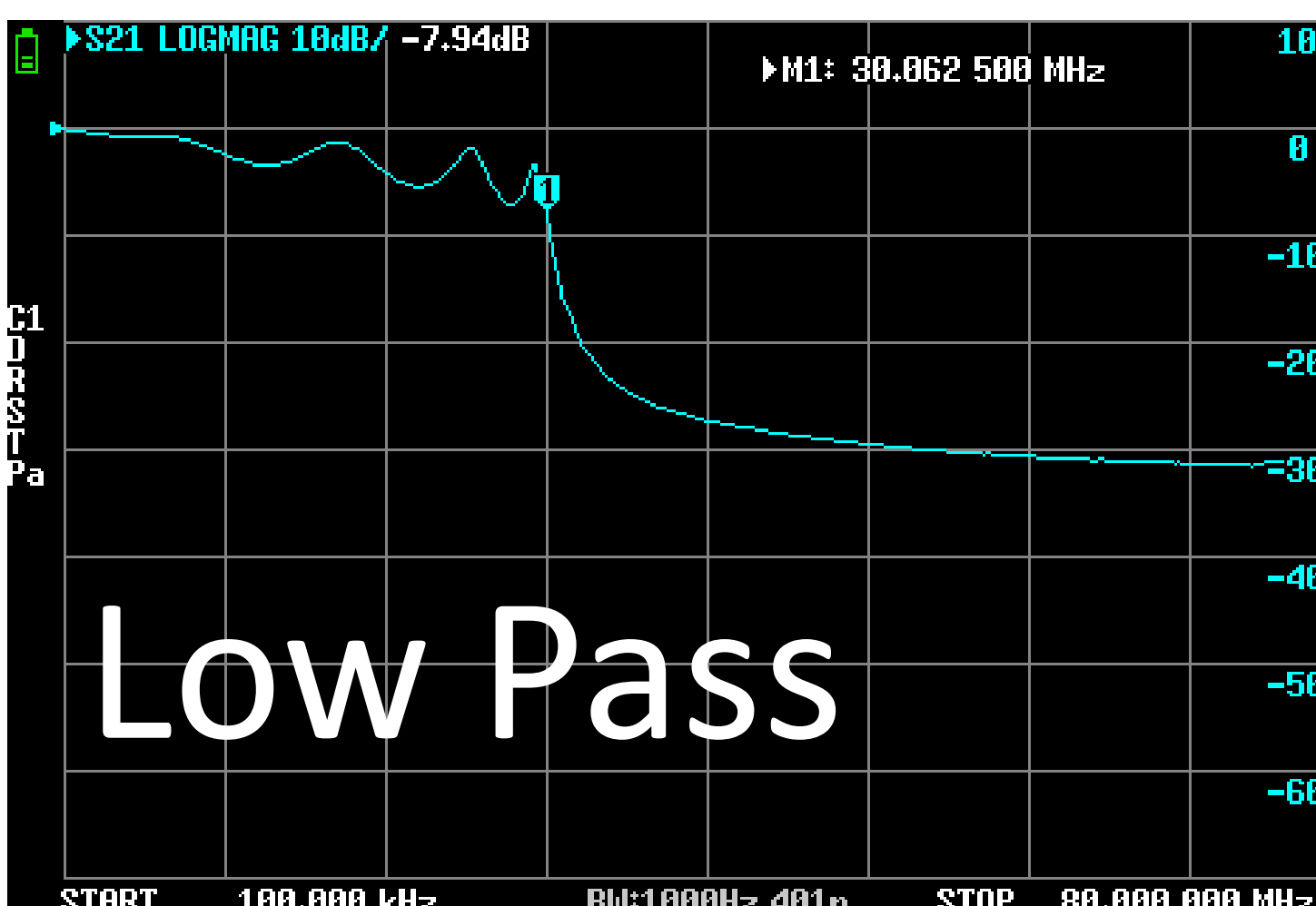
Results



Band Pass Filter Frequency Response Test Results:



Low Pass Filter Frequency Response Test Results:



Results - Analysis

- Between an input power of 0 to 6 W, the amplifier circuit showed a consistent power output curve for both the forward and reflected power (and by extension the total power). The average gain between 0 to 6 W was recorded as **13.89 dB**. Note, that between an input power of 1 W and 6 W, the gain fell from **15.24 dB** (ratio of 33.43) to **13.14 dB** (ratio of 20.165), with an average gain of **14.92 dB** in this range.
- Between an input power of 6 to 7 W, the gain of the amplifier increased from **13.15 dB** (ratio of 20.165), to **14.16 dB** (ratio of 26.08).
- Between an input power of 7 to 9 W, an average gain of **13.74 dB** was recorded.
- At 10 W of output power, the gain fell to **12.43 dB** (ratio of 17.5).
- The observed inconsistency in the gain at higher input transmit powers is likely due to the Xiegu X6100 transceiver used in testing, which is known to offer inconsistent output powers above 6-7 W, particularly being noted to struggle to output a clean 10 W signal. Alternatively, at 10 W, the amplifier could be experiencing saturation, leading to a poor drive response.
- However, despite some slight transceiver related oddities, the amplifier circuit performed excellently, with an overall mean average gain of **14.60 dB** between 0 to 10 W being recorded.
- Low pass filter** - frequency response was almost perfect, showing an expected Chebyshev reply and frequency cutoff occurring just past the peak of the last ripple.
- Band pass filter** - showed an excellent, sharp cutoff in the high pass filter portion of the frequency response. In the low pass portion of the spectrum, the attenuation was a little more gradual between 30 – 36 MHz before more rapidly falling off, but displayed acceptable results for the performance levels needed. However, there is a noticeable ripple in the pass band, and while it doesn't severely limit amateur radio bands, it may cause slightly reduced performance on 15 and 17 meter amateur bands.

Conclusions

- The amplifier exhibited an average gain of 14.60 dB, offering a substantial power boost over the standard radio, increasing the transmission range, as well as an improved signal strength on the receiving end, leading to improved readability.
- The filters attenuate the frequencies they were expected to, with the LPF showing perfect results, and the BPF showing workable attenuation with some caveats.
- The cooling features provided exceptional performance, with the amp experiencing no overheating issues in testing.
- The final amplifier prototype cost was approx. \$589.25, with a mass of 1.504 kg and a volume of 5670 cm³. The values for cost and mass were all within the specifications set out for this project (less than \$600 and 3 kg), with the volume being larger, exceeding the 3000 cm³ specification, although the amplifier offered excellent gain performance to offset this increased size.
- In conclusion, this project designed a functional, portable HF amplifier that could be paired with a low power HF radio for emergency communication use during natural disasters.

Future Work

- To continue amplifier development, it is intended to implement a built-in power/SWR meter to further improve versatility.
- The filter layout of the bandpass filter will be iterated upon to flatten the pass band, and an investigation into the viability of implementing an antenna tuner feature will be conducted.

References

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